

Chapter 7

Residue Theorem

Analytic function

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Cauchy-Riemann condition

⇓

Cauchy Theorem

⇓

Cauchy Integral

⇓

Taylor Expansion, Laurent Expansion

⇓

Residue Theorem

Suppose we have two singularities, then $\frac{1}{(z-1)(z-2)}$

$$\oint_c f(z) dz = 2\pi i [a_{-1}^{(1)} + a_{-1}^{(2)}]$$

$$f(z) = \sum_{n=-\infty}^{\infty} a^{(1)} (z - b_1)^n$$

residue: the coefficient of $(z - b_1)^{-1}$

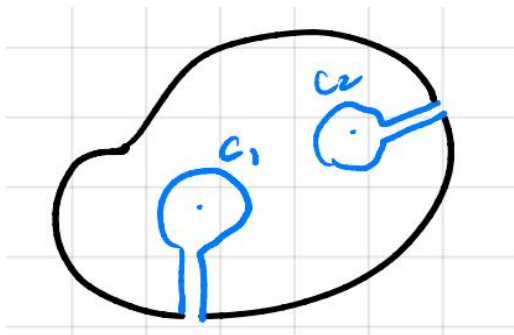
$$\text{res}f(b_1) \triangleq a_{-1}^{(1)}$$

Then find $\text{res}f(b_2)$

$$\sum_{n=-\infty}^{\infty} a^{(2)} (z - b_2)^n$$

Then we can get $a_{-1}^{(2)}$

Proof



$$\oint_{C_{all}} f(z) dz = 0$$

$$\oint_C f(z) dz = \oint_{c_1} f(z) dz + \oint_{c_2} f(z) dz$$

$$\oint_{c_1} f(z) dz = \oint_{c_1} \sum_{n=0}^{\infty} a_n (z - b_1)^n dz = \sum_{n=-\infty}^{\infty} \oint_{c_1} a_n (z - b_1)^n dz = 2\pi i a_{-1}^{(1)}$$

Likewise,

$$\oint_{c_2} f(z) dz = 2\pi i a_{-1}^{(2)}$$

□

$$\oint_C (z - b_1)^n dz = \begin{cases} 0 & n \neq -1 \\ 2\pi i & n = -1 \end{cases}$$

$$\oint_c f(z)dz = 2\pi i \sum_{k=1}^n \text{res}f(b_k)$$

$b_1, b_2, \dots, b_k$ are singularities

How to find a_{-1}

1. Do Laurent Expansion

2. first order pole

$$\lim_{z \rightarrow b} f(z)(z - b) = a_{-1}$$

$$f(z)(z - b)|_{z=b} = a_{-1}$$

Example $\frac{1}{z^2+1}$

$$\frac{1}{z^2+1} = \frac{1}{(z+i)(z-i)}$$

$$b_1 = i$$

$$\frac{1}{(z+i)(z-i)} \cdot (z-i)|_{z=i} = \frac{1}{2i} = \text{res}$$

$$b_2 = -i$$

$$\frac{1}{(z+i)(z-i)} \cdot (z+i)|_{z=-i} = \frac{1}{-2i} = \text{res}$$

Example $\frac{e^{iaz} - e^{ibz}}{z^2}$

$$\lim_{z \rightarrow 0} \frac{e^{iaz} - e^{ibz}}{z^2} \cdot z = \lim_{z \rightarrow 0} \frac{e^{iaz} - e^{ibz}}{z} = i(a - b)$$

If $f(z) = \frac{P(z)}{Q(z)}$

$$a_{-1} = \lim_{z \rightarrow b} (z - b)f(z) = \lim_{z \rightarrow b} (z - b) \frac{P(z)}{Q(z)} = \frac{P(b)}{Q'(b)}$$

3. m-th order pole

(1) . 2-nd order

$$f(z) = a_{-2}(z - b)^{-2} + a_{-1}(z - b)^{-1} + a_0 + a_1(z - b) + \dots$$

$$(z - b)^2 f(z) = a_{-2} + a_{-1}(z - b)^{-1} + \dots$$

$$\frac{d}{dz} [(z - b)^2 f(z)]|_{z=b}$$

(2) . 3-rd order

$$(a - b)^3 f(z) = a_{-3} + a_{-2}(z - b) + a_{-1}(z - b)^2 + \dots$$

$$2! \times a_{-1} = \frac{d^2}{dz^2} [(z - b)^3 f(z)]|_{z=b}$$

(3) . m-th order

$$a_{-1} = \frac{1}{(m - 1)!} \frac{d^{m-1}}{dz^{m-1}} [(z - b)^m f(z)]|_{z=b}$$

Example

$$f(z) = \frac{1}{(z - 1)^2(z - 2)(z - 3)}$$

$$\lim_{z \rightarrow 2} (z - 2) \cdot f(z) = \frac{1}{1^2 \cdot (-1)} = -1$$

$$\lim_{z \rightarrow 3} (z - 3) \cdot f(z) = \frac{1}{2^2 \cdot 1} = \frac{1}{4}$$

$$a_{-1} = \frac{d}{dz} (f(z)(z - 1)^2)|_{z=1}$$

Section 6.2

$$I = \int_0^{2\pi} R(\sin\theta, \cos\theta) d\theta$$

R means *rational* (有理)

$$R(x) = \frac{P(x)}{Q(x)}$$

We can change x into $\cos\theta, \sin\theta$

Example

$$\frac{\cos^2\theta + 2\sin\theta\cos\theta + 1}{\sin^2\theta + 2\cos^2\theta + 2}$$

which is rational and trigonometric

Example find $\int_0^\pi \frac{1}{1+\epsilon\cos\theta} d\theta$

Suppose z is on the unit circle.

$$z = e^{i\theta}$$

$$\sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i} \quad \cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

On the unit circle, so we can get,

$$\cos\theta = \frac{z + z^{-1}}{2} = \frac{z^2 + 1}{2z}$$

$$\sin\theta = \frac{z^2 - 1}{2iz}$$

$$dz = ie^{i\theta} d\theta \quad d\theta = \frac{1}{iz} dz$$

Then we can get,

$$\begin{aligned} & \int_0^\pi \frac{1}{1 + \epsilon\cos\theta} d\theta \\ &= \frac{1}{2} \int_0^\pi \frac{1}{1 + \epsilon\cos\theta} d\theta = \frac{1}{2} \oint_{|z|=1} \frac{1}{1 + \epsilon \cdot \frac{z+1}{2z}} \frac{1}{iz} dz \\ &= \frac{1}{i} \oint_{|z|=1} \frac{dz}{2z + \epsilon(z^2 + 1)} dz \\ &= \frac{1}{i} \oint_{|z|=1} \frac{1}{\epsilon z^2 + 2z + \epsilon} dz \end{aligned}$$

$$\oint_C f(z) dz = 2\pi i \sum_{k=1}^K \text{res}(b_k)$$

We can get the residues.

$$\frac{-2 \pm \sqrt{4 - 4\epsilon^2}}{2\epsilon} : b_1 \quad b_2$$

If $|\epsilon| < 1$

Take $\epsilon = 0.5$ as an example.

$$\text{Integral} = 2\pi i \cdot \frac{1}{\epsilon(z - b_1)(z - b_2)} (z - b_2)|_{z=b_2} = 2\pi i \frac{1}{\epsilon(b_1 - b_2)}$$

Then,

$$\int_0^\pi \frac{1}{1 + \epsilon \cos \theta} d\theta = \frac{2\pi}{\sqrt{1 - \epsilon^2}}$$

Example

$$\begin{aligned} & \int_0^\pi \frac{d\theta}{1 + \sin^2 \theta} \\ &= \frac{1}{2} \int_0^{2\pi} \frac{d\theta}{1 + \sin^2 \theta} \\ &= \frac{1}{2} \oint_{|z|=1} \frac{1}{1 + \left(\frac{z - z^{-1}}{2i}\right)^2} \frac{1}{iz} dz \\ &= \frac{1}{2i} \oint_{|z|=1} \frac{-4z}{z^4 - 6z^2 + 1} dz \\ &= \frac{1}{2i} \oint_{|z|=1} \frac{-4z}{(z - b_1)(z - b_2)(z - b_3)(z - b_4)} dz \\ &= \frac{1}{2i} \cdot 2\pi i \sum_{k=1}^M \text{res} f(n_k) \end{aligned}$$

Example

$$\begin{aligned}
& \int_0^{2\pi} \cos^{2n} \theta d\theta \\
&= \oint_{|z|=1} \left(\frac{z + z^{-1}}{2} \right)^{2n} \frac{1}{iz} dz \\
&= \frac{1}{i} \oint_{|z|=1} \frac{(z^2 + 1)^{2n}}{z^{2n} z^{2n+1}} dz \\
&= \frac{1}{i} \frac{1}{2^{2n}} \cdot \oint_{|z|=1} \frac{(z^2 + 1)^{2n}}{z^{2n} \cdot z^{2n+1}} dz
\end{aligned}$$

$z = 0$ is the pole of order $2n + 1$

$$\begin{aligned}
a_{-1} &= \frac{1}{(m-1)!} \frac{d^{m-1}}{dz^{m-1}} (z-b)^m f(z) \Big|_{z=b} \\
&= \frac{1}{(2n)!} \frac{d^{2n}}{dz^{2n}} [(z^2 + 1)^{2n}] \Big|_{z=0}
\end{aligned}$$

Let $\int_0^1 g(z) = (z^2 + 1)^{2n}$, we can get

$$g'(z) = 2n(z^2 + 1)^{2n-1} \cdot 2z$$

Then

$$g^{(2n)}(z) = (2n)! \cdot (2z)^{2n}$$

Example

$$\int_0^{2\pi} \frac{dx}{(a + b \cos \theta)^2}$$

